

GSE98923 co-expression analysis
CEMiTool baseline - 54-sample balanced design

2026-09-25

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1 Scope

This report documents the computational analysis of the balanced 54-sample subset of GEO accession **GSE98923** used in the CEMiTool baseline workflow.

The report is intended as an analysis record and potential supplementary document for a scientific manuscript. Large machine-readable matrices are not printed in full in the document; complete files are preserved separately and indexed in the supplementary manifest.

2 Experimental design

The analysis contains **54 samples**, covering two cultivars, three developmental stages, three years, and three biological replicates per cultivar-stage-year combination.

Table 1: Sample counts by cultivar, developmental stage, and year.

Cultivar	Stage	Year	Freq
Cabernet Sauvignon	FruitSet	2012	3
Pinot noir	FruitSet	2012	3
Cabernet Sauvignon	Harvest	2012	3
Pinot noir	Harvest	2012	3
Cabernet Sauvignon	Veraison	2012	3
Pinot noir	Veraison	2012	3
Cabernet Sauvignon	FruitSet	2013	3
Pinot noir	FruitSet	2013	3
Cabernet Sauvignon	Harvest	2013	3
Pinot noir	Harvest	2013	3
Cabernet Sauvignon	Veraison	2013	3
Pinot noir	Veraison	2013	3
Cabernet Sauvignon	FruitSet	2014	3
Pinot noir	FruitSet	2014	3
Cabernet Sauvignon	Harvest	2014	3
Pinot noir	Harvest	2014	3
Cabernet Sauvignon	Veraison	2014	3
Pinot noir	Veraison	2014	3

3 Expression data

The processed matrix used for CEMiTool contains **29971 genes** and **54 samples**.

Expression values were derived from the processed RPKM matrix distributed with GSE98923 and transformed as:

$$\log_2(\text{RPKM} + 1)$$

The complete matrices are available under:

- data/processed/expression_rpkms.tsv
- data/processed/expression_log2rpkms.tsv

4 CEMiTool analysis

CEMiTool identified **11 modules** in the baseline analysis.

The analysis used Pearson correlation, an unsigned network, a signed TOM, a minimum module size of 30, and a fixed random seed of 1234.

Automatic beta selection did not identify a usable value in the initial baseline run. To reproduce the established 54-sample baseline, the soft-thresholding power was fixed explicitly at **beta = 10** using `set_beta = 10` and `force_beta = FALSE`. This value is specific to the present expression matrix and should be re-evaluated if the sample set, normalization, or filtering changes. The recorded beta for this run is **10**.

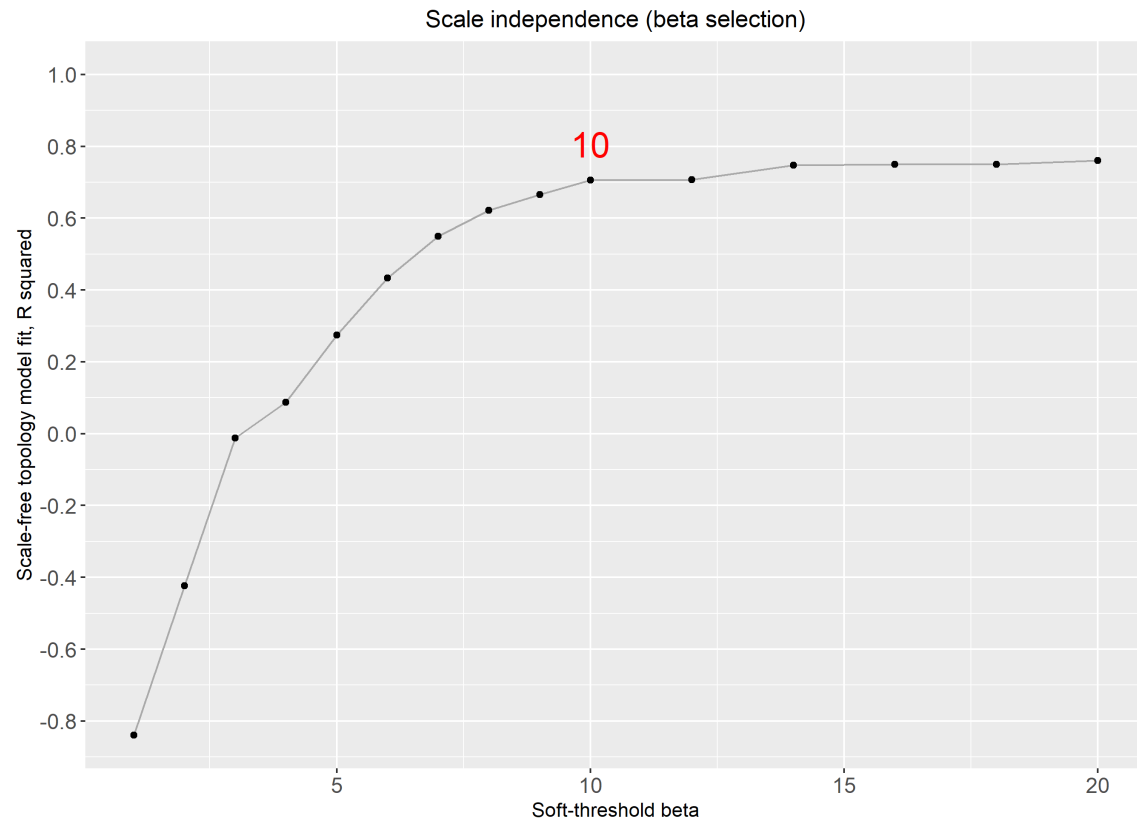
Table 2: Recorded analysis parameters.

parameter	value
dataset	GSE98923
samples	54
transform	log2(RPKM+1)
correlation	pearson
network_type	unsigned
tom_type	signed
beta_strategy	manual_set_beta_beta10
set_beta	10
force_beta	FALSE
beta	10
r2	0.706374247063269
filter	TRUE
filter_pval	0.1
min_module_size	30
seed	1234
modules	11

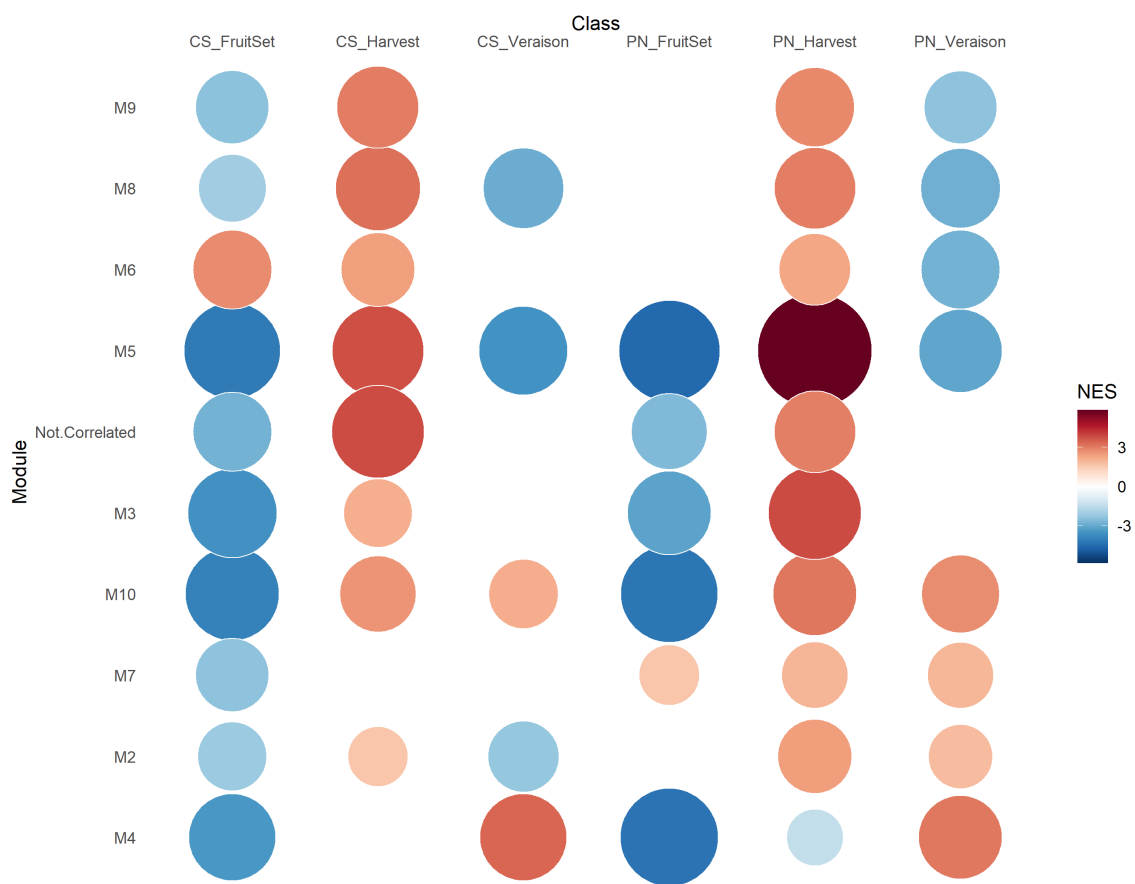
5 Diagnostic and analysis figures

All figures generated by the analysis are stored in `results/figures/`. PDF figures are converted to PNG copies for reliable inclusion in Word and HTML reports.

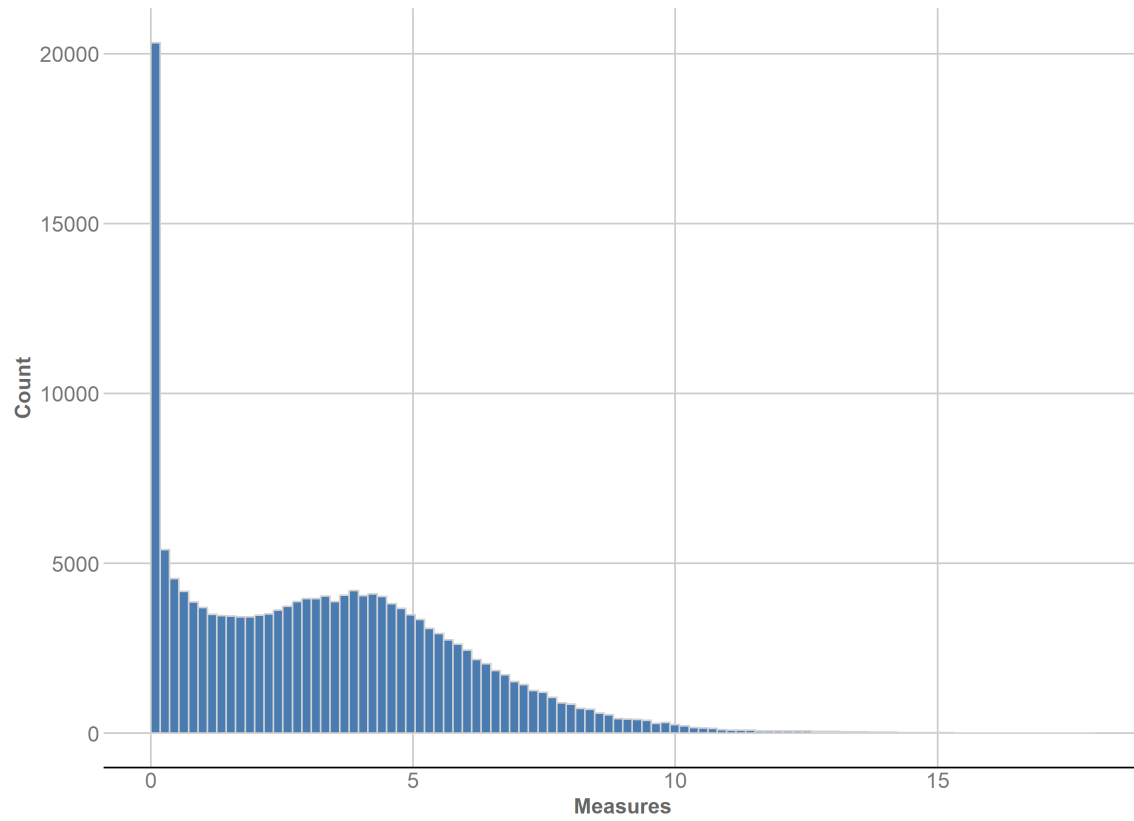
5.1 beta_r2



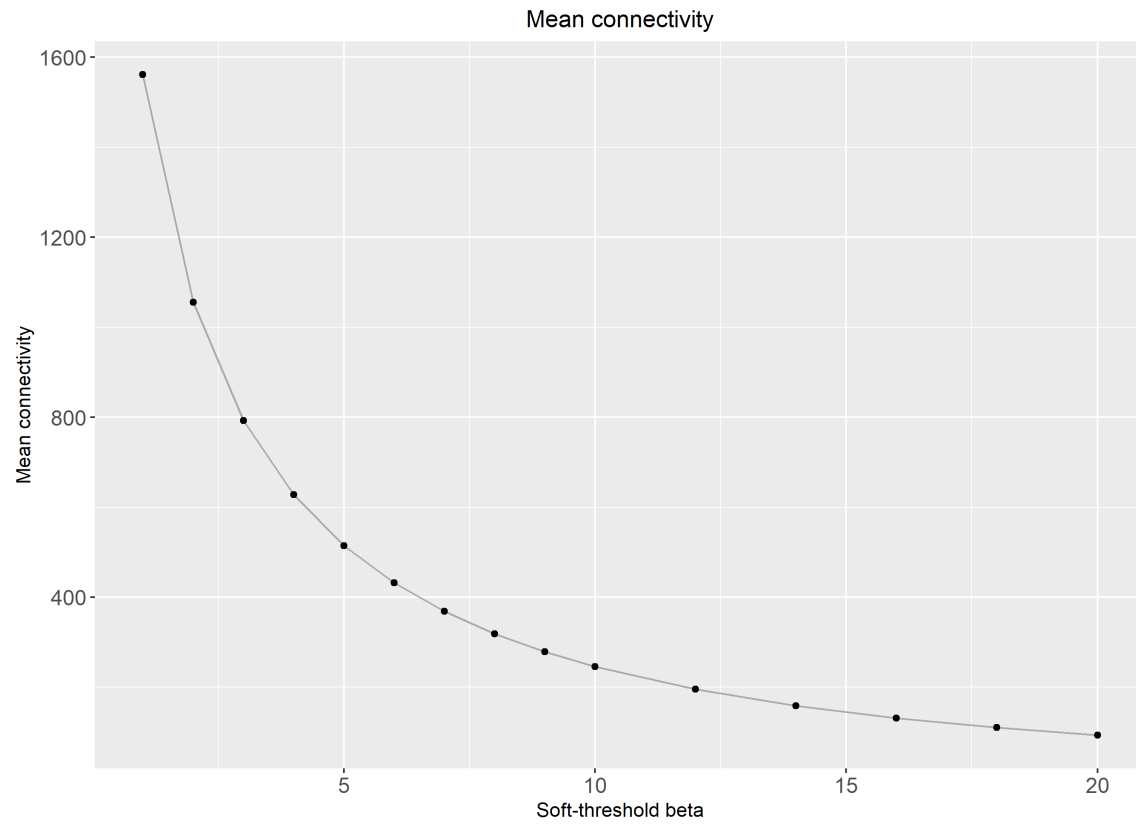
5.2 gsea



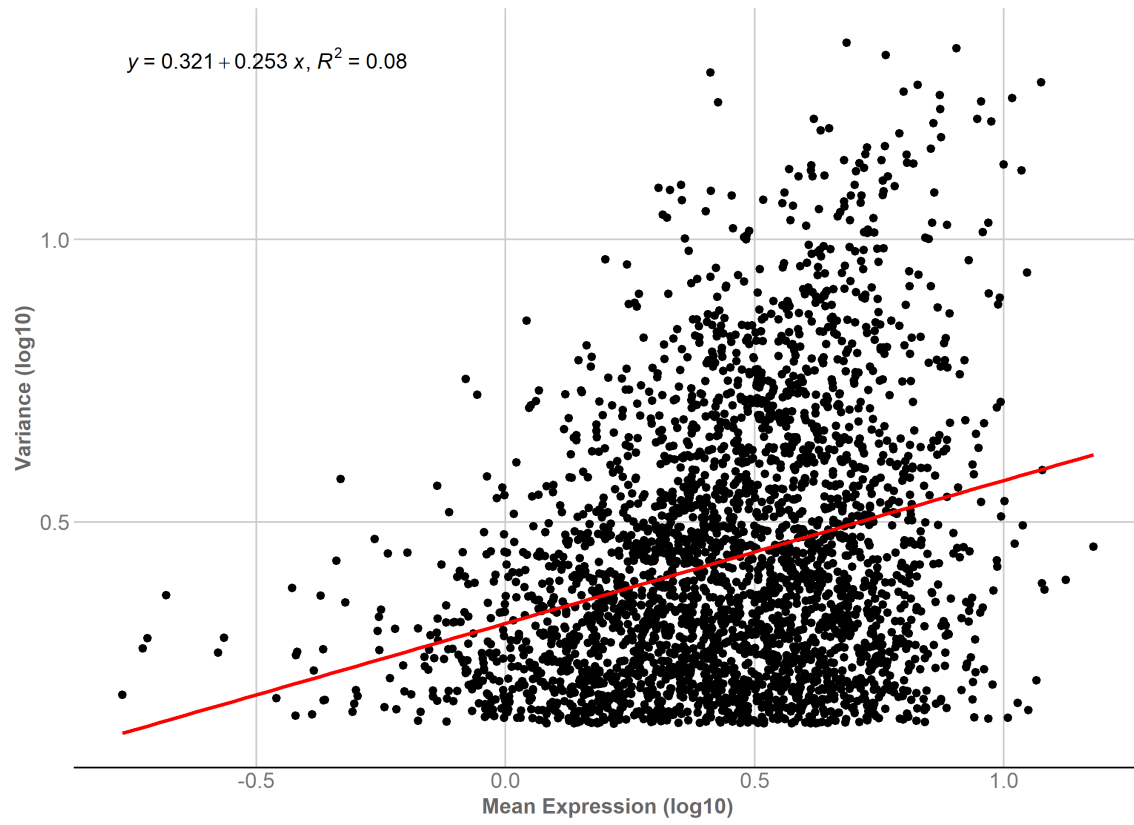
5.3 hist



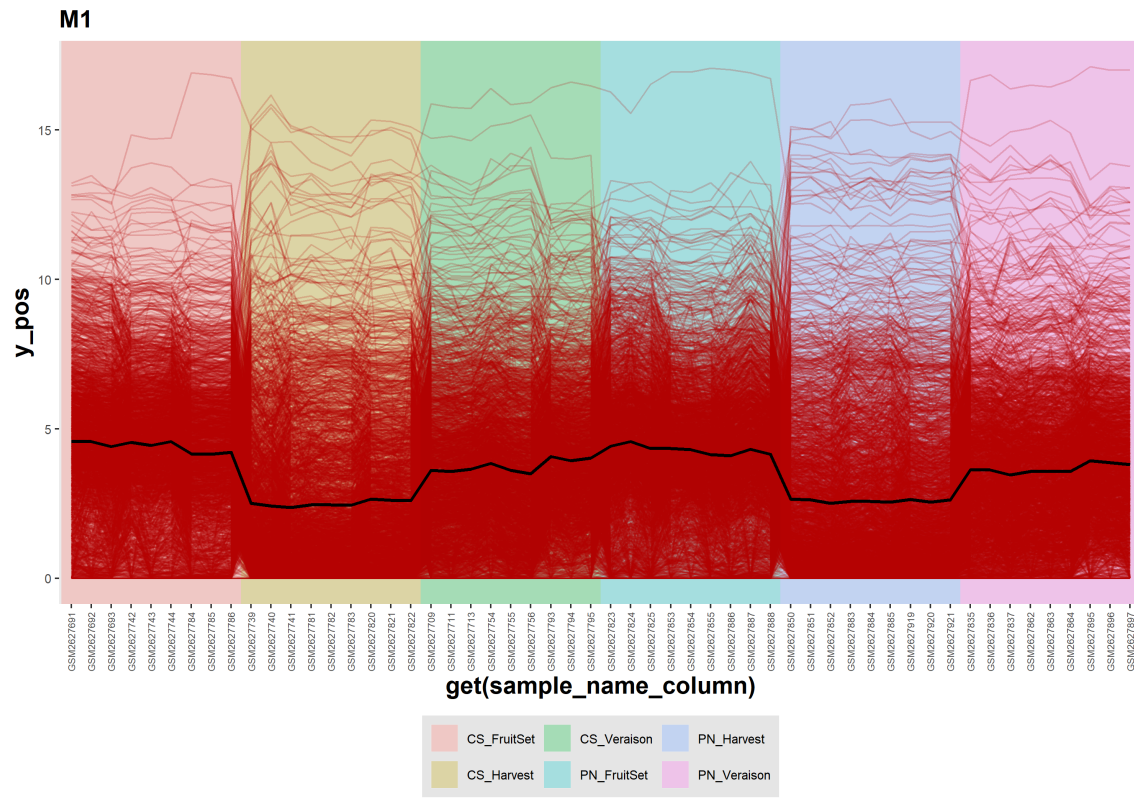
5.4 mean_k



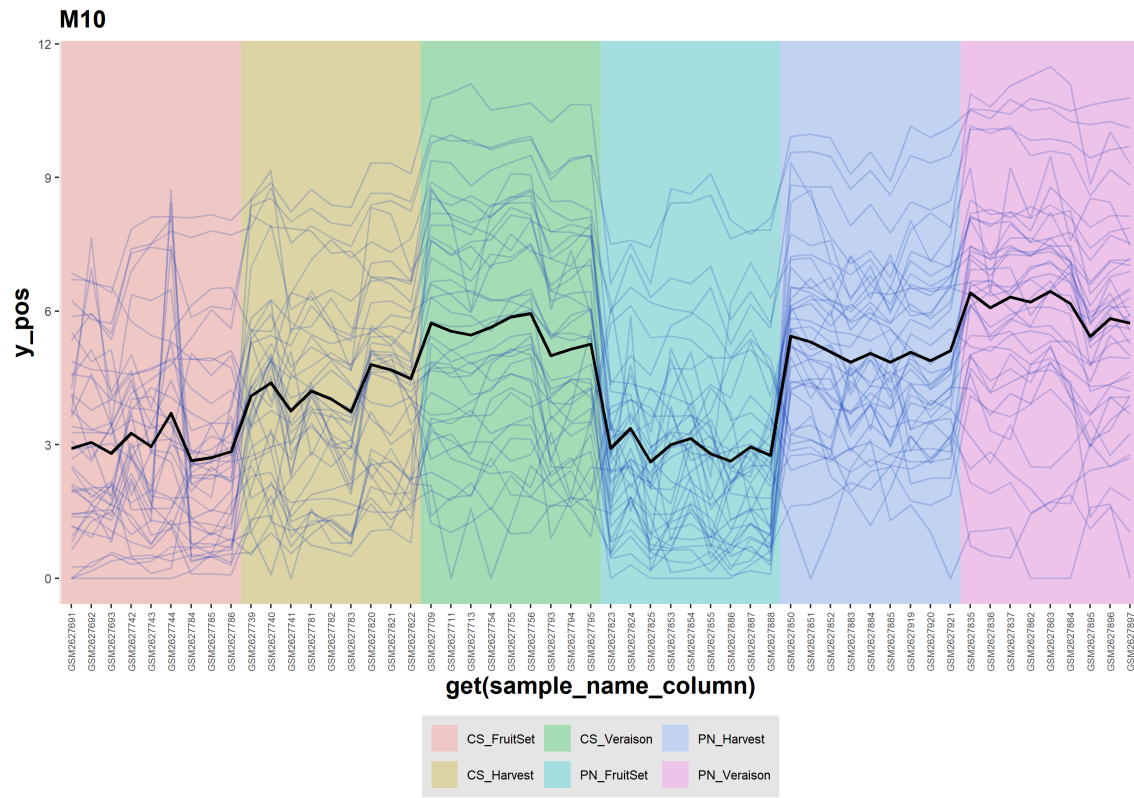
5.5 mean_var



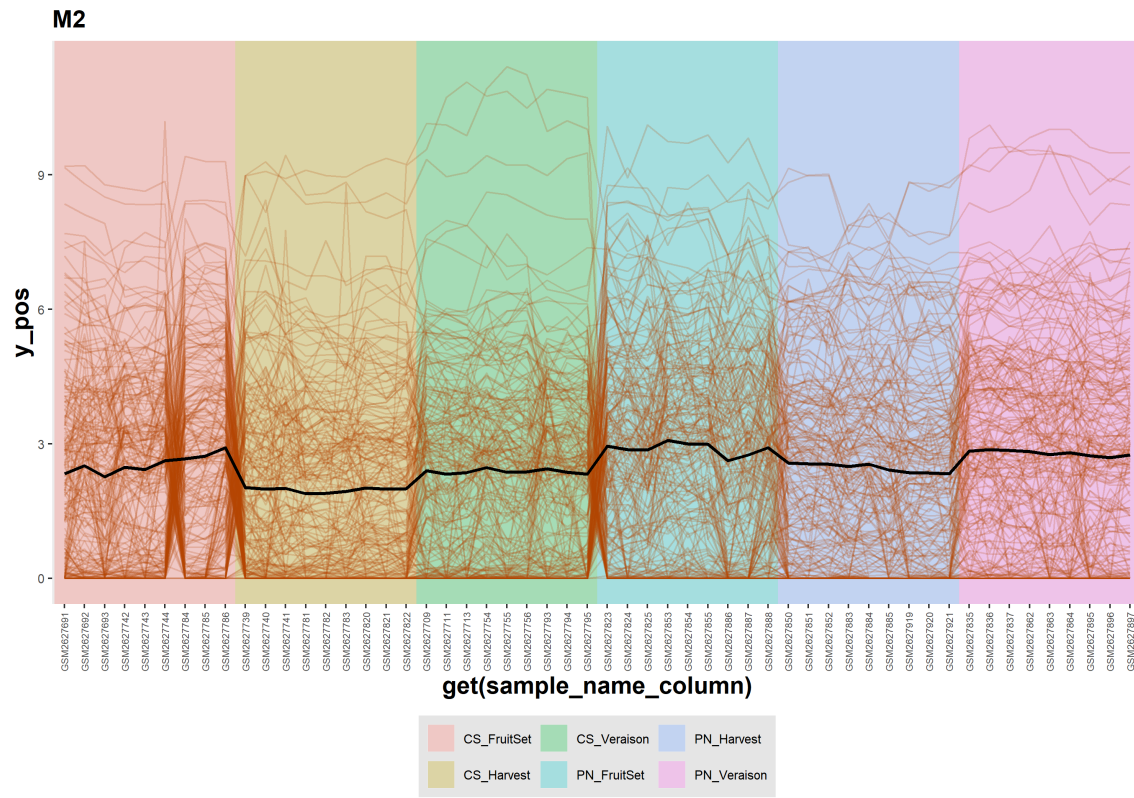
5.6 profile_M1



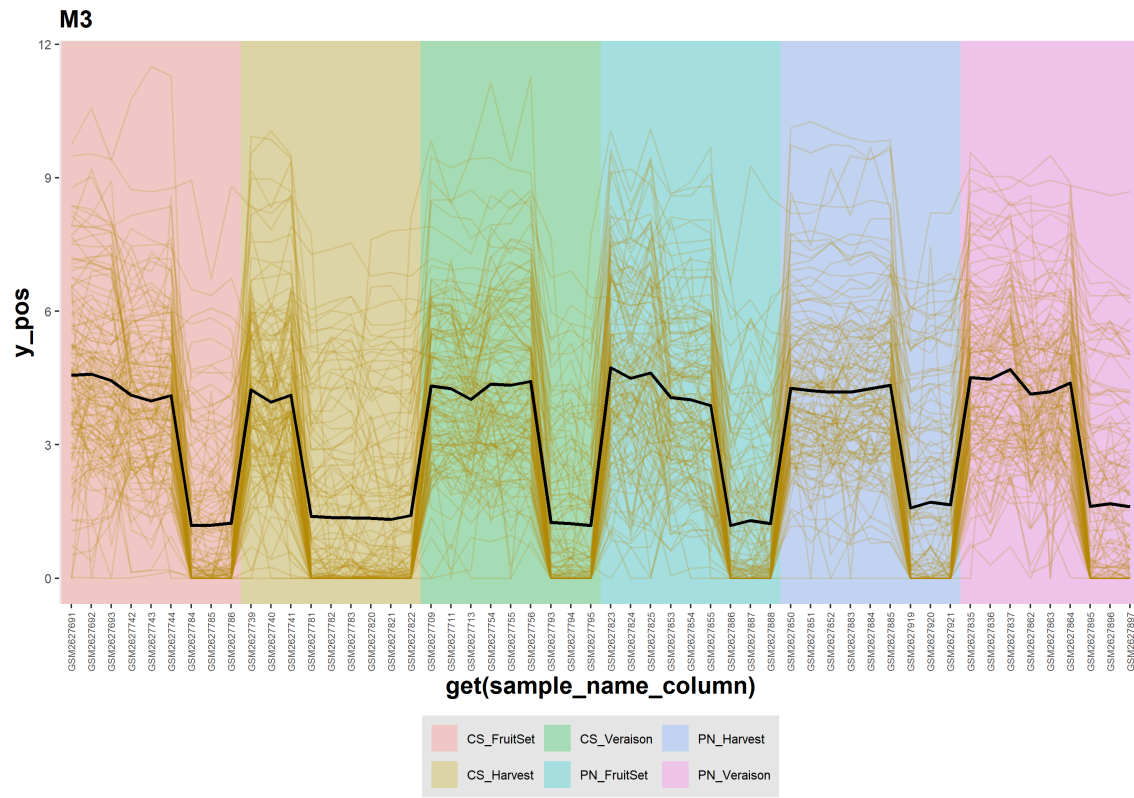
5.7 profile_M10



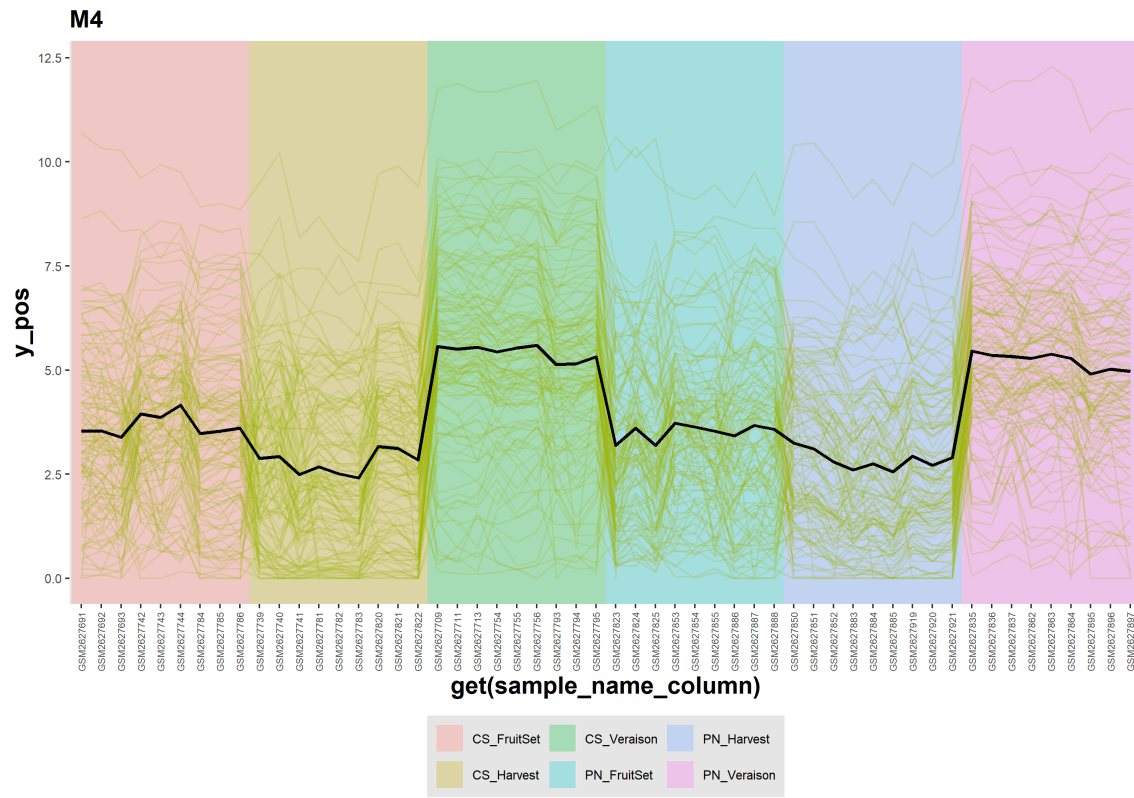
5.8 profile_M2



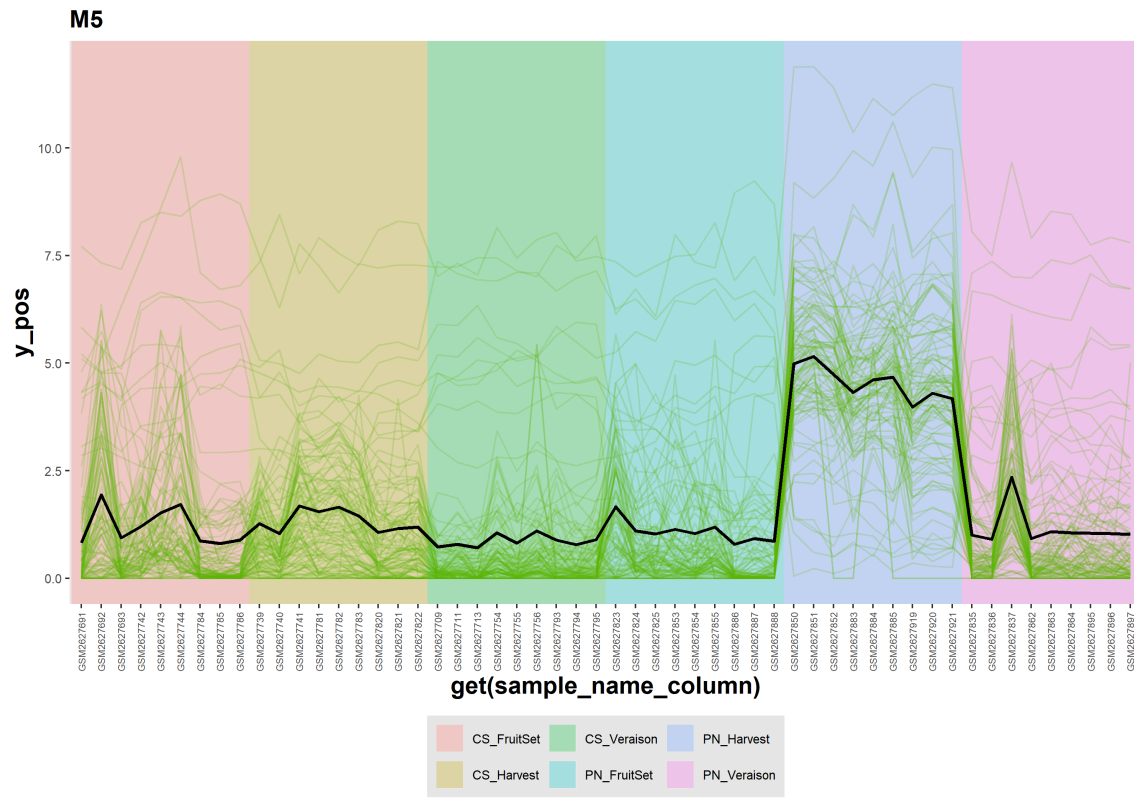
5.9 profile_M3



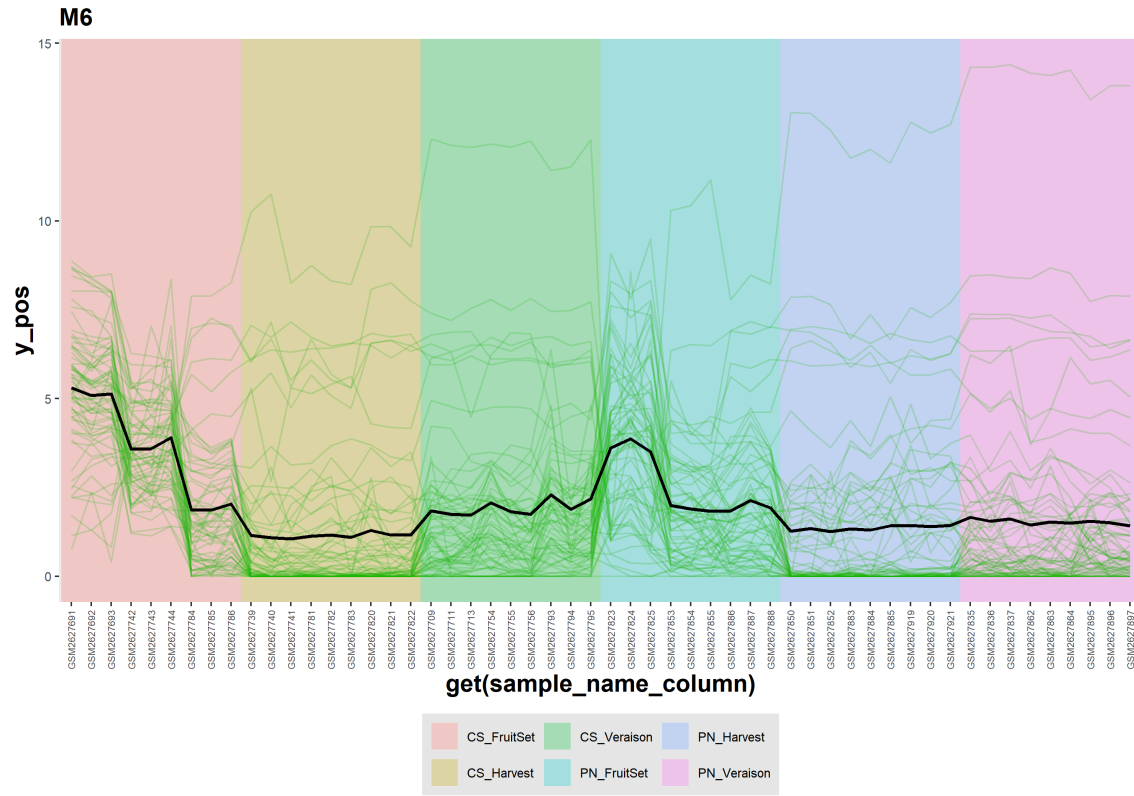
5.10 profile_M4



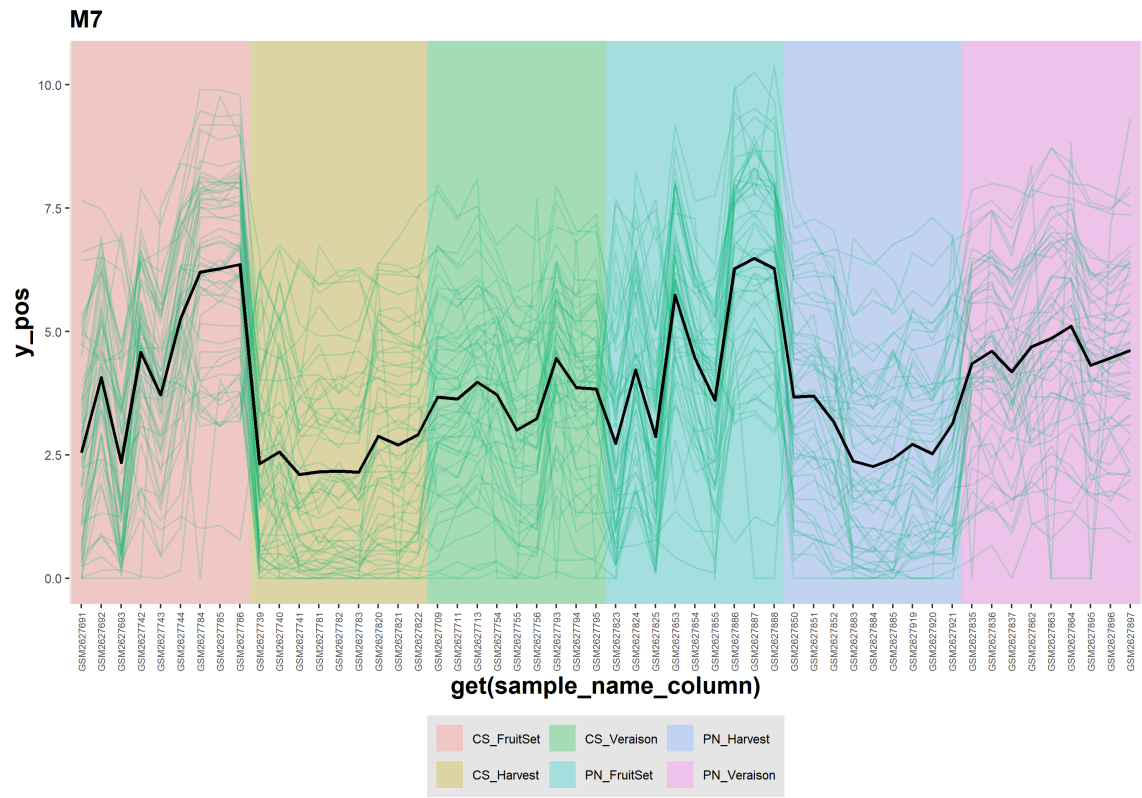
5.11 profile_M5



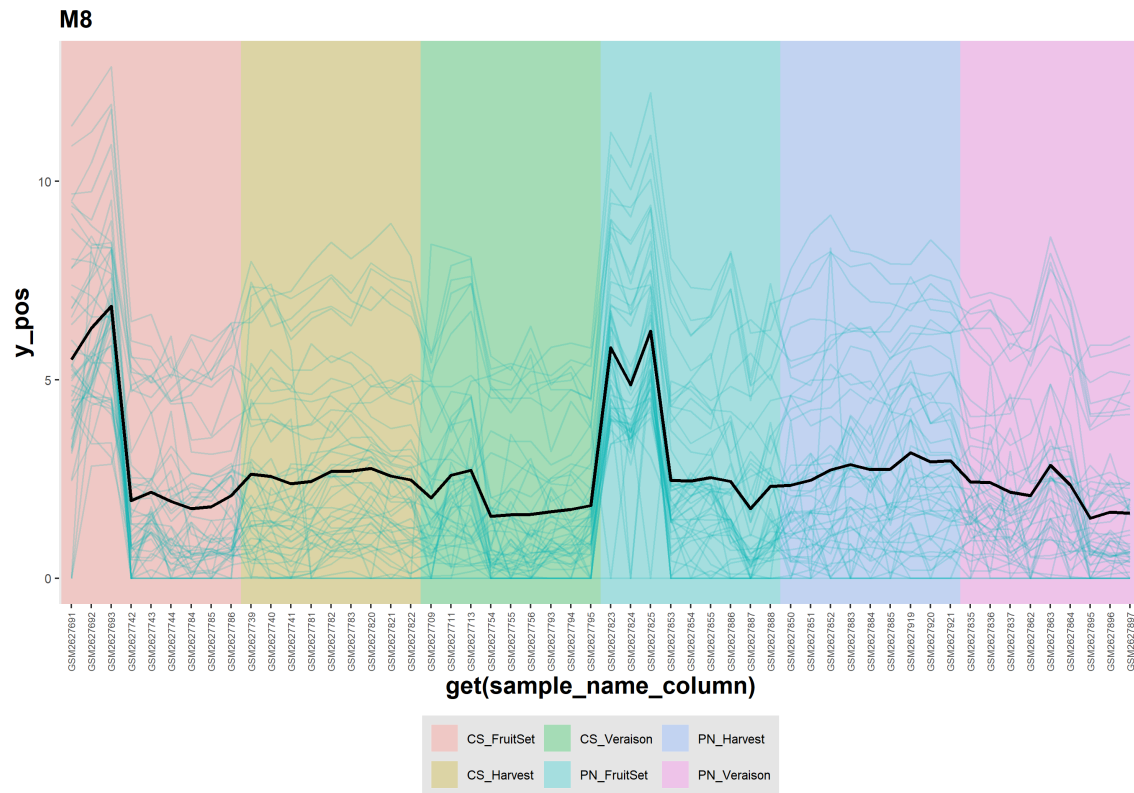
5.12 profile_M6



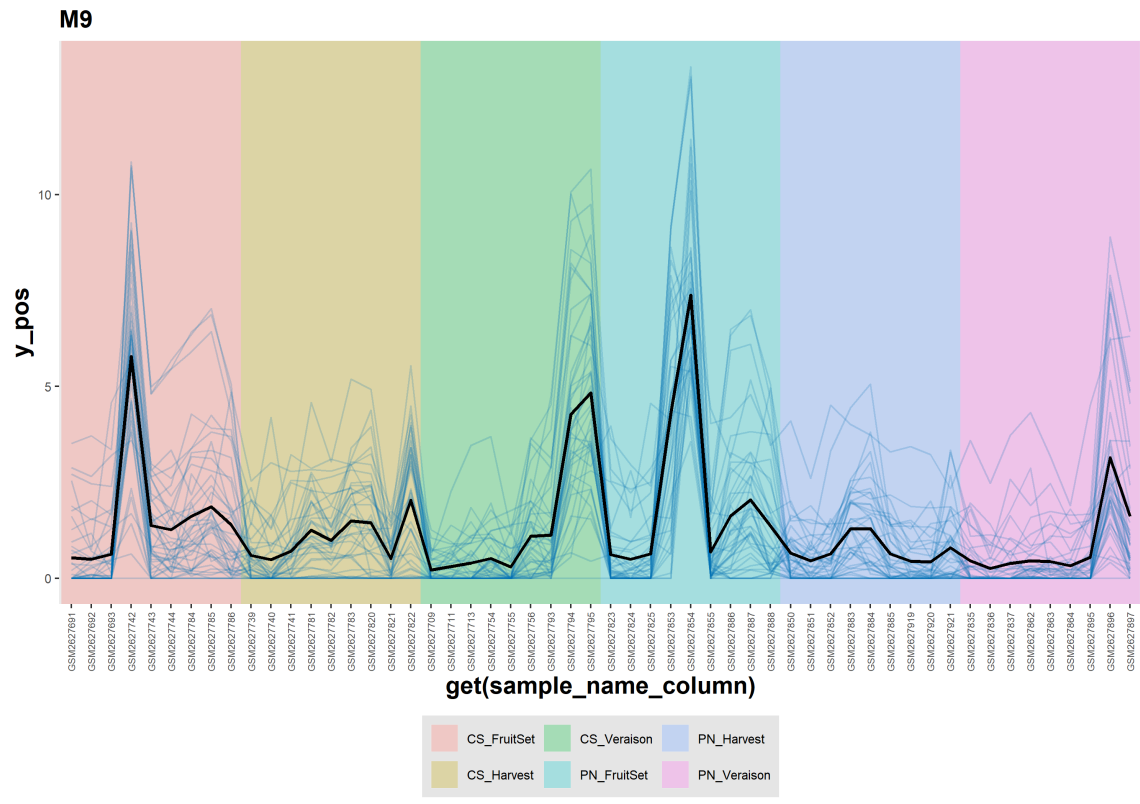
5.13 profile_M7



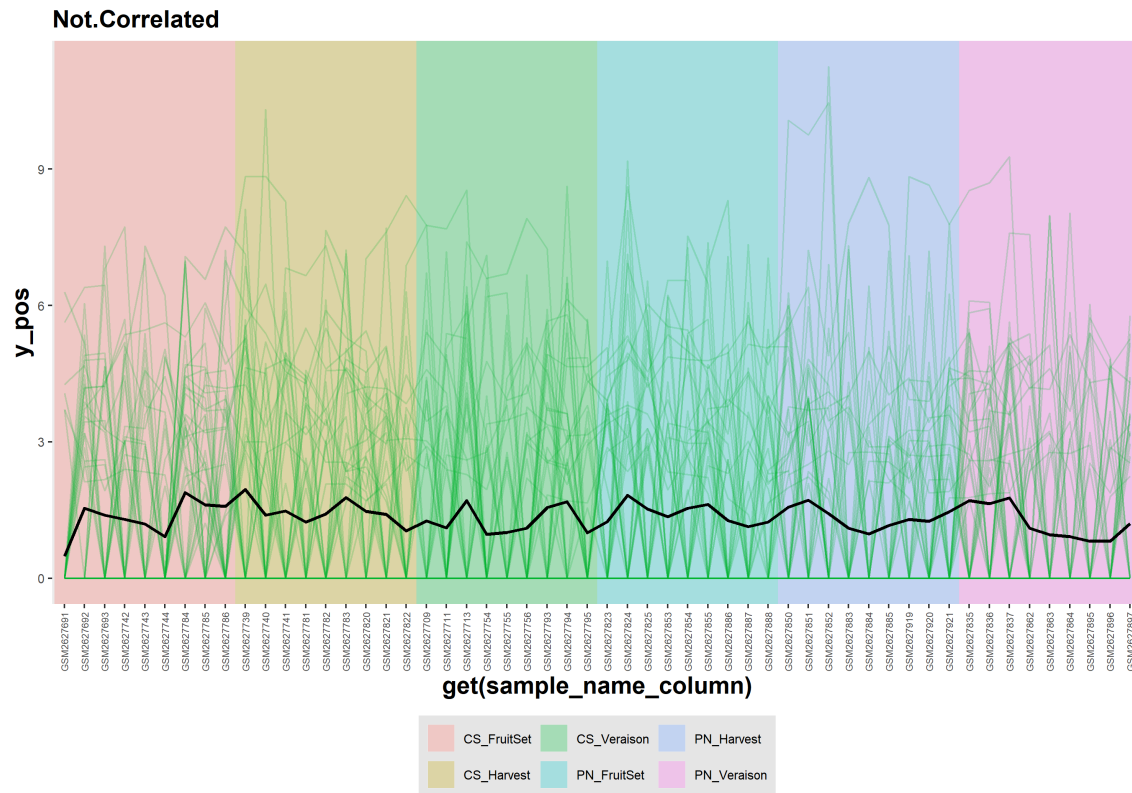
5.14 profile_M8



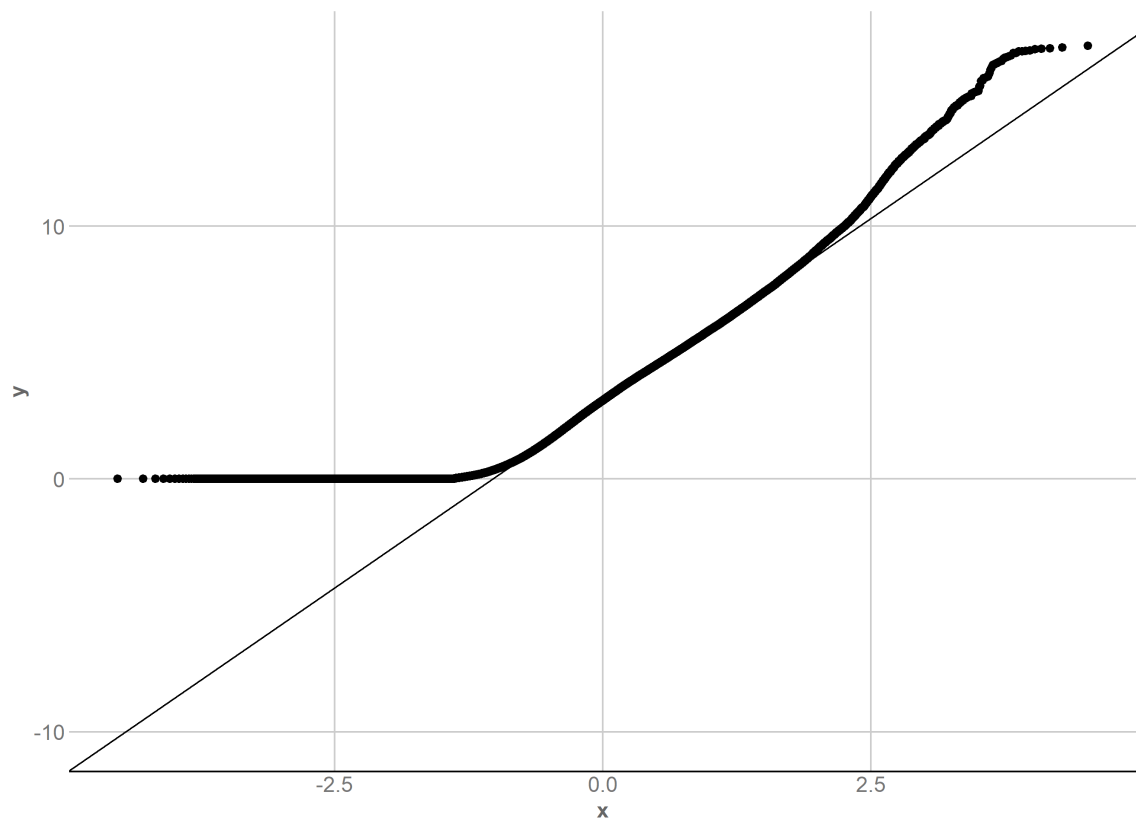
5.15 profile_M9



5.16 profile_Not_Correlated



5.17 qq



6 Exported result tables

The following section provides a compact view of every tabular result exported by CEMiTool. Complete tables remain available as machine-readable files under `results/tables/`.

6.1 `beta_fit_indices.tsv`

Dimensions: 15 rows x 10 columns.

Table 3: `beta_fit_indices.tsv`

Power	SFT	R.sq	slope	truncated.R.sq	mean.k.	median.k.	max.k.	Density	Centralization	Heterogeneity
1	0.8394009	0.5707687	0.7946939	1561.857117	54.661221	111.9801	0.5122523	0.1805457	0.3207976	
2	0.4232012	0.3873307	0.5948254	1055.584872	0.787817	0.721103	0.462069	0.2133693	0.4790910	
3	0.0123211	0.0444688	0.7254709	792.106958	84.928451	453.8582	0.259792	0.2171812	0.5801171	
4	0.0875404	-	0.8310439	627.739736	75.701751	1274.2760	0.2058838	0.2121878	0.6544427	
			0.1174076							
5	0.2744513	-	0.8129080	514.611285	28.426901	136.6584	0.1687803	0.2041506	0.7137761	
			0.2192085							

	Power	SFT.R.sq	slope	truncated.R.sq	mean.k.	median.k.	max.k.	Density	Centralization	Heterogeneity
6	0.4333524	-	0.7362054	431.828944	21.508541	1026.4451	0.1416297	0.1951480	0.7635877	
	0.3044837									
7	0.5495793	-	0.6942176	368.657523	39.871359	35.4799	0.120911	0.1860264	0.8068294	
	0.3627214									
8	0.6217077	-	0.6648029	318.950732	77.028728	58.7238	0.1046083	0.1771490	0.8452677	
	0.4267637									
9	0.6650263	-	0.6414873	278.909912	28.173737	92.8474	0.0914759	0.1686699	0.8800415	
	0.4781301									
10	0.7063742	-	0.6502824	246.049851	89.373377	35.5400	0.0806985	0.1606465	0.9119266	
	0.5251829									
12	0.7067470	-	0.6230659	195.571181	34.480706	40.3966	0.0641427	0.1459880	0.9690884	
	0.6147726									
14	0.7472853	-	0.6797739	158.919239	7.11593	564.3466	0.0521218	0.1330578	1.0196729	
	0.6734742									
16	0.7496827	-	0.6990696	131.367807	2.38920	502.0230	0.0430855	0.1216459	1.0654044	
	0.7344417									
18	0.7495778	-	0.7234977	110.099295	3.69994	449.9688	0.0361100	0.1115423	1.1073859	
	0.7871557									
20	0.7605758	-	0.7500230	93.33028	40.81002	405.8351	0.0306101	0.1025615	1.1463678	
	0.8268108									

6.2 enrichment_es.tsv

Dimensions: 10 rows x 7 columns.

Table 4: enrichment_es.tsv

pathway	CS_FruitSet	CS_Veraison	CS_Harvest	PN_FruitSet	PN_Veraison	PN_Harvest
M10	-0.7230231	0.5985374	0.4662943	-0.7631498	0.7632057	0.5953022
M2	-0.2702467	-0.4045229	0.1897572	0.1528338	0.4009491	0.3168241
M3	-0.4928964	-0.0775830	0.2777637	-0.4016256	0.2979030	0.5528034
M4	-0.4797068	0.7662442	-0.2166083	-0.5704144	0.7178583	-0.2388118
M5	-0.5946456	-0.6852390	0.5099956	-0.6279539	-0.5903679	0.8708191
M6	0.5981419	-0.2428080	0.3700474	0.1467911	-0.5620152	0.3800916
M7	-0.4028705	-0.2620536	-0.2969209	0.3406279	0.5067387	0.3471849
M8	-0.3513185	-0.6778466	0.5233773	0.3072853	-0.6231626	0.5444628
M9	-0.4249585	-0.1595745	0.5184045	0.3130987	-0.5479842	0.5255984
Not.Correlated	-0.4570197	-0.1856715	0.6155754	-0.4134418	-0.2873009	0.5145339

6.3 enrichment_nes.tsv

Dimensions: 10 rows x 7 columns.

Table 5: enrichment_nes.tsv

pathway	CS_FruitSet	CS_Veraison	CS_Harvest	PN_FruitSet	PN_Veraison	PN_Harvest
M10	-3.959396	2.1492587	2.615432	-4.2536734	2.728041	3.118858
M2	-2.141531	-2.2761001	1.614254	0.9429160	1.839075	2.451360
M3	-3.541564	-0.4093924	2.099724	-3.1346970	1.304286	3.856798
M4	-3.403567	3.3899077	-1.230025	-4.3302647	3.115328	-1.424578
M5	-4.156126	-3.5193848	3.767544	-4.5950476	-3.083399	5.848706
M6	2.777735	-1.1341715	2.424789	0.7430314	-2.774107	2.283311
M7	-2.394055	-1.1530162	-1.444617	1.6434296	1.941217	1.979208
M8	-2.044511	-2.9146050	3.236747	1.4406157	-2.837919	2.984385
M9	-2.405142	-0.6670530	3.002033	1.4350954	-2.385699	2.803384
Not.Correlated	-2.773385	-0.8343909	3.831869	-2.5856587	-1.359322	2.973914

6.4 enrichment_padj.tsv

Dimensions: 10 rows x 7 columns.

Table 6: enrichment_padj.tsv

pathway	CS_FruitSet	CS_Veraison	CS_Harvest	PN_FruitSet	PN_Veraison	PN_Harvest
M10	0.0000000	0.0000225	0.0000024	0.0000000	0.0000000	0.0000000
M2	0.0000000	0.0000000	0.0004024	0.6718346	0.0000035	0.0000000
M3	0.0000000	1.0000000	0.0000024	0.0000000	0.0774487	0.0000000
M4	0.0000000	0.0000000	0.1552106	0.0000000	0.0000000	0.0377113
M5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
M6	0.0000000	0.3151261	0.0000022	0.8589894	0.0000000	0.0000089
M7	0.0000037	0.3151261	0.0707547	0.0165577	0.0001668	0.0011522
M8	0.0002388	0.0000000	0.0000000	0.0829238	0.0000000	0.0000000
M9	0.0000254	1.0000000	0.0000000	0.0829238	0.0000047	0.0000001
Not.Correlated	0.0000000	1.0000000	0.0000000	0.0000025	0.0732555	0.0000000

6.5 module.tsv

Dimensions: 3050 rows x 2 columns.

Table 7: module.tsv - first 40 rows shown

genes	modules
VIT_19s0014g02740	M1
VIT_08s0056g01610	M1
VIT_12s0059g00590	M1
VIT_01s0010g02030	M1
VIT_08s0056g01600	M1
VIT_05s0049g00520	M1

genes	modules
VIT_02s0025g03390	M1
VIT_11s0016g00210	M1
VIT_16s0022g00960	M1
VIT_07s0005g00660	M1
VIT_19s0090g01370	M1
VIT_06s0004g03240	M1
VIT_06s0004g04650	M1
VIT_05s0049g00760	M1
VIT_17s0000g03280	M4
VIT_19s0014g02730	M1
VIT_01s0011g05110	M6
VIT_03s0180g00070	M1
VIT_00s0187g00160	M1
VIT_18s0001g14270	M1
VIT_19s0090g01340	M1
VIT_09s0002g05180	M1
VIT_07s0031g02210	M1
VIT_19s0090g01000	M1
VIT_03s0038g01510	M1
VIT_16s0022g01770	M1
VIT_05s0094g00340	M1
VIT_18s0001g14260	M1
VIT_02s0025g01050	M1
VIT_19s0090g01350	M1
VIT_02s0033g01330	M1
VIT_02s0033g01120	M10
VIT_04s0008g05010	M1
VIT_02s0025g03600	M1
VIT_02s0025g04330	M1
VIT_13s0019g04280	M1
VIT_10s0003g00090	M1
VIT_03s0038g02520	M1
VIT_11s0016g04920	M1
VIT_12s0028g01060	M1

The complete table is provided as a supplementary machine-readable file.

6.6 parameters.tsv

Dimensions: 11 rows x 2 columns.

Table 8: parameters.tsv

Parameter	Value
filter	TRUE
apply_vst	FALSE
filter_pval	0.1
cor_method	pearson
min_ngen	30
merge_similar	TRUE
diss_thresh	0.8
r2	0.706374247063269
beta	10
phi	0.500166177070297
n_mods	11

6.7 selected_genes.txt

Dimensions: 3049 rows x 1 columns.

Table 9: selected_genes.txt - first 40 rows shown

VIT_19s0014g02740
VIT_08s0056g01610
VIT_12s0059g00590
VIT_01s0010g02030
VIT_08s0056g01600
VIT_05s0049g00520
VIT_02s0025g03390
VIT_11s0016g00210
VIT_16s0022g00960
VIT_07s0005g00660
VIT_19s0090g01370
VIT_06s0004g03240
VIT_06s0004g04650
VIT_05s0049g00760
VIT_17s0000g03280
VIT_19s0014g02730
VIT_01s0011g05110
VIT_03s0180g00070
VIT_00s0187g00160
VIT_18s0001g14270
VIT_19s0090g01340
VIT_09s0002g05180
VIT_07s0031g02210
VIT_19s0090g01000
VIT_03s0038g01510

VIT_19s0014g02740

VIT_16s0022g01770
VIT_05s0094g00340
VIT_18s0001g14260
VIT_02s0025g01050
VIT_19s0090g01350
VIT_02s0033g01330
VIT_02s0033g01120
VIT_04s0008g05010
VIT_02s0025g03600
VIT_02s0025g04330
VIT_13s0019g04280
VIT_10s0003g00090
VIT_03s0038g02520
VIT_11s0016g04920
VIT_12s0028g01060
VIT_18s0001g12830

The complete table is provided as a supplementary machine-readable file.

6.8 summary__eigengene.tsv

Dimensions: 11 rows x 55 columns.

The in-document preview is omitted because this matrix has 55 columns. The complete table remains available as a machine-readable file.

6.9 summary__mean.tsv

Dimensions: 11 rows x 55 columns.

The in-document preview is omitted because this matrix has 55 columns. The complete table remains available as a machine-readable file.

6.10 summary__median.tsv

Dimensions: 11 rows x 55 columns.

The in-document preview is omitted because this matrix has 55 columns. The complete table remains available as a machine-readable file.

7 Supplementary data inventory

The full analysis files are indexed in `reports/supplementary/manifest.tsv`. The manifest records relative paths, file sizes, and MD5 checksums.

Table 10: Supplementary file inventory.

file	bytes	md5
data/metadata/phenotypes.tsv	1296	3ded458ab03ff3e1694c33db594c19f9
data/metadata/sample_geo_map.tsv	10654	c0e7260e2115a3fb85dd76228b953c44
data/metadata/sample_images.tsv	1497	2f2700c2e90786b44e078b0cfabdbfa7
data/metadata/samples.tsv	9957	494e1078fc9806589f3024eed79c8136
data/processed/expression_log2rpkm.rds	9656439	77f6b173e329b9c1583d1bd52c0ab5ca
data/processed/expression_log2rpkm.tsv	21802281	95761a42392a04d55642f378fe0fa5ce
data/processed/expression_rpkm.tsv	11158928	636a4103dc8b388862503c5c989b3690
logs/parameters.txt	288	42e700fbf2f7fdcc032db19fdaf38e28
logs/run_summary.txt	147	153756a512a8a707d6406da5dcc2d0d2
logs/session_info.txt	5832	46cfc2f7438b3a095a050a648622de80
reports/cemitoool/report.html	4053372	72edf3164422e6cec1d29d62d4baf4b2
reports/diagnostics/diagnostics.html	975533	02876519ac07af41cab568e017076094
reports/native_report_status.tsv	103	cb9c1034c7cf2e211fb29ce01b16b718
results/figures/beta_r2.pdf	5718	83cfaa117a752d16bf6f42a5a5f22e79
results/figures/gsea.pdf	9481	a8e0c3c482e0c878231afe9227f971dc
results/figures/hist.pdf	5105	b5dafc083d17fd9d9eafecf8f9908d19
results/figures/mean_k.pdf	5468	450d829952c18a11db0e7b08df03f015
results/figures/mean_var.pdf	173388	2b4a703d540829622eb8b7a2ff99b6cd
results/figures/profile.pdf	715657	59f97246e8cd190a556c579099cc90a5
results/figures/qq.pdf	1392120	5f384f61a0ac5630906baf5453ba76d1
results/figures/sample_tree.pdf	6798	3f42183cbcd26905c0771ac939acc8d4
results/tables/beta_fit_indices.tsv	2527	6ec616e12b6c75d209918578a4a717cb
results/tables/enrichment_es.tsv	1273	5aaa94fc1ac921d3b7ecb34dea12834d
results/tables/enrichment_nes.tsv	1213	70f2139cabd0bf1825dbb315982b389f
results/tables/enrichment_padj.tsv	1336	57ff84dcd2d32ab4d7c3616426701ebd
results/tables/module.tsv	80042	c1f78a7ab02f444690e3c6ca9778dcdb
results/tables/modules_genes.gmt	72623	76a41959c5260da575318d5dbe1621f8
results/tables/parameters.tsv	247	31188bc3cad8c270fbc0a1877f93af2
results/tables/selected_genes.txt	57950	3fbe5f3efd8f2a6f69851ea0df1e0ad7
results/tables/summary_eigengene.tsv	12093	dc0ea4013ab26493d6dc8cb8bc7bfd89
results/tables/summary_mean.tsv	10893	b9b1595a8ca08e7e003592a0c1ad22ee
results/tables/summary_median.tsv	10071	ec27d56bf7f2c816bdb63d71004c66bd

8 Reproducibility

R version 4.6.1 (2026-06-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
LAPACK version 3.12.1

```

locale:
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    LC_MONETARY=Spanish_Chile.utf8
[4] LC_NUMERIC=C                  LC_TIME=Spanish_Chile.utf8

time zone: America/Santiago
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods    base

other attached packages:
[1] CEMiTool_1.36.0 GEOquery_2.80.0 Biobase_2.72.0 BiocGenerics_0.58.1 generics_0.1.4

loaded via a namespace (and not attached):
[1] splines_4.6.1 ggplotify_0.1.3 R.oo_1.27.1 tibble_3.3.1
[5] ggpp_0.6.1 polyclip_1.10-7 preprocessCore_1.74.0 enrichit_0.2.1
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9 Notes for manuscript use

This document is a computational analysis report, not a substitute for biological interpretation. Statements about biological mechanism, causal relationships, novelty, and comparison with previous literature should be supported separately with appropriate experimental evidence and verified references.

10 Post-CEMiTool module statistics

This section is an additional statistical layer performed after construction of the beta-10 CEMiTool network. It is not part of the native CEMiTool report and is not stored inside the CEMiTool object.

10.1 Network calibration and sensitivity

Beta 10 is the primary network and beta 7 is retained as a sensitivity analysis. Module labels are arbitrary across runs, so robustness is evaluated using shared genes rather than matching labels by name.

Table 11: Best beta7-to-beta10 matches based on gene membership.

Module_beta7	Module_beta10	Shared_genes	Jaccard	Percent_beta7_retained
M1	M1	2150	0.912	91.9
M2	M2	203	0.864	90.6
M3	M3	105	0.761	93.8
M4	M5	101	0.894	95.3
M5	M4	91	0.705	92.9
M6	M8	46	0.630	64.8
M7	M10	34	0.723	81.0
M8	M9	40	0.952	100.0
Not.Correlated	Not.Correlated	17	0.298	100.0

Module_beta7	Module_beta10	Shared_genes	Jaccard	Percent_beta7_retained
--------------	---------------	--------------	---------	------------------------

10.2 Module eigengene model

For each beta-10 biological module, PC1 of standardized module-gene expression was used as the module eigengene. PC1 sign was oriented to correlate positively with mean module expression.

Model:

```
Eigengene ~ Cultivar * Stage + Year
```

Type III ANOVA used sum-to-zero contrasts and Benjamini-Hochberg FDR.

10.3 FDR overview by module

Table 12: FDR by module and model effect.

	Module	FDR_Cultivar	FDR_Stage	FDR_Year	FDR_Cultivar:Stage
1	M1	0.4943000	0.0000000	0.0031450	0.039650
5	M10	0.0000028	0.0000000	0.1165000	0.001052
9	M2	0.0000000	0.0002871	0.0052300	0.019510
13	M3	0.0136800	0.0070290	0.0000000	0.019510
17	M4	0.1448000	0.0000000	0.7054000	0.884000
21	M5	0.0000000	0.0000000	0.0011460	0.000000
25	M6	0.0401300	0.0000000	0.0031450	0.100400
29	M7	0.0732200	0.0000001	0.0009819	0.540700
33	M8	0.5370000	0.0005303	0.0006119	0.972900
37	M9	0.6657000	0.0954400	0.0205000	0.567300

10.4 Significant Cultivar x Stage interactions

5 of 10 modules passed $FDR < 0.05$. Strongest: **M5** ($FDR=5.29e-22$).

Table 13: Significant Cultivar x Stage interactions.

Module	F value	p_value	FDR
M5	190.926286	5.29e-23	5.29e-22
M10	10.235546	2.10e-04	1.05e-03
M2	5.562952	6.86e-03	1.95e-02
M3	5.402947	7.81e-03	1.95e-02
M1	4.274767	1.98e-02	3.97e-02

A significant interaction means that the Cabernet-versus-Pinot difference changes with developmental stage. It does not by itself demonstrate a causal relationship with skin thickness.

10.5 Cabernet Sauvignon vs Pinot noir within stage

The contrast is Cabernet Sauvignon minus Pinot noir. Negative estimates therefore indicate a lower module eigengene in Cabernet Sauvignon.

Table 14: Significant cultivar contrasts within stage.

Module	Stage	estimate	Direction	p.value	FDR_global
M5	Harvest	-22.279	Cabernet Sauvignon < Pinot noir	2.85e-28	8.56e-27
M2	Veraison	-25.367	Cabernet Sauvignon < Pinot noir	7.98e-16	1.20e-14
M2	Harvest	-21.832	Cabernet Sauvignon < Pinot noir	1.28e-13	1.28e-12
M2	FruitSet	-15.558	Cabernet Sauvignon < Pinot noir	2.40e-09	1.80e-08
M10	Harvest	-3.356	Cabernet Sauvignon < Pinot noir	1.79e-07	1.07e-06
M3	Harvest	-6.989	Cabernet Sauvignon < Pinot noir	7.62e-05	3.81e-04
M10	Veraison	-2.128	Cabernet Sauvignon < Pinot noir	3.18e-04	1.36e-03
M6	FruitSet	6.146	Cabernet Sauvignon > Pinot noir	2.40e-03	8.99e-03
M5	Veraison	-2.458	Cabernet Sauvignon < Pinot noir	8.78e-03	2.93e-02

10.6 Year effect

Year passed FDR < 0.05 in **8 of 10 modules**.

Table 15: Modules with significant Year effect.

	Module	p_value	FDR
3	M1	1.89e-03	3.14e-03
11	M2	3.66e-03	5.23e-03
15	M3	1.75e-21	1.75e-20
23	M5	4.59e-04	1.15e-03
27	M6	1.64e-03	3.14e-03
31	M7	2.95e-04	9.82e-04
35	M8	1.22e-04	6.12e-04
39	M9	1.64e-02	2.05e-02

This supports keeping 2012, 2013 and 2014 explicit. The completed robustness layer below tests whether the Cultivar x Stage pattern is stable across years.

10.7 Interpretation boundaries

- GSE98923 is berry pericarp, not isolated skin.
- There is no direct skin-thickness phenotype in this dataset.
- Only two cultivars are compared.
- Several modules have a significant Year effect.
- The baseline uses the historical GEO RPKM matrix.

The subsequent layers use $\text{Eigengene} \sim \text{Cultivar} * \text{Stage} * \text{Year}$, functional annotation, hub prioritization, skin-only external comparison and modern raw-read reprocessing. They remain observational and do not convert module association into phenotype causality.

11 Integrated evidence from T-004 through T-008

This section connects the vintage, functional, candidate-gene, external-validation and modern-processing layers. Each statement is derived from the canonical machine-readable files loaded below. The historical beta-10 network remains unchanged.

11.1 T-004: robustness across vintages

The full model was $\text{Eigengene} \sim \text{Cultivar} * \text{Stage} * \text{Year}$. M10 met the prespecified module-level reproducibility rule. M5, M2, M3 and M1 were year-dependent under at least one criterion, but M5 Harvest and M2 Veraison/Harvest retained the same Cabernet-minus-Pinot direction and passed global FDR in all three years. M3 Harvest was dominated in magnitude by 2013, and M1 was not established as stable.

Table 16: Vintage-robustness classification of the five interaction modules.

Module	Classification	Repeated stage	Single-year dominant	Sign reversal
M5	year-dependent	Harvest	Veraison	FALSE
M10	reproducible	Veraison,Harvest		FALSE
M2	year-dependent	Veraison,Harvest		TRUE
M3	year-dependent	Harvest		TRUE
M1	year-dependent		FruitSet	TRUE

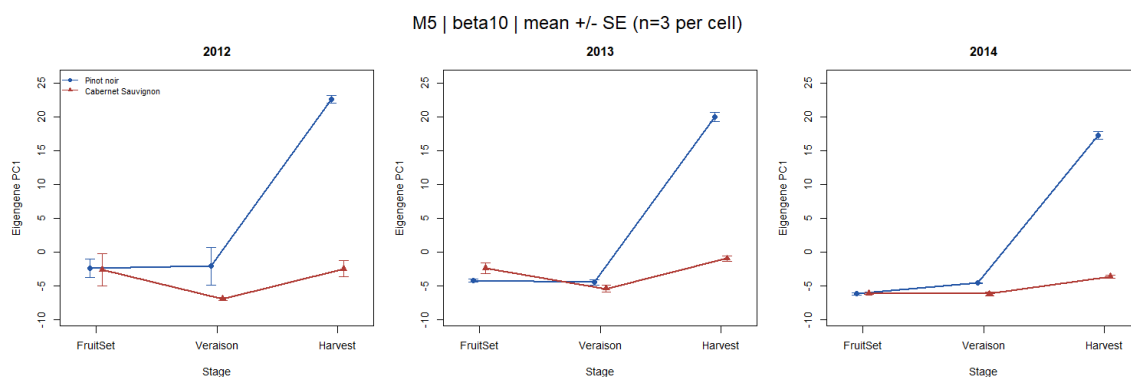


Figure 1: M5 observed eigengene means and biological-replicate standard errors by stage, cultivar and year. Inferential contrasts come from the full factorial model.

11.2 T-005 and T-005A: functional enrichment and ontology audit

MapMan v3 was the primary annotation layer because it covered 2,608 of 3,050 beta-10 genes. M5 was globally enriched for phenolic/stilbenoid and PAL categories. Newer v5.1 labels assigned several of the same chromosome-16 genes to CHS/flavonoid categories, so the exact CHS-versus-STS identity remains unresolved.

The official 2026-07-26 Gene Ontology audit identified 390 obsolete IDs among 3,556 represented Grapedia terms. After filtering, the annotated background remained 519 genes and all six global-FDR GO hits occurred in M9. Sparse coverage prevents treating the absence of GO hits in M5, M10, M2, M3 or M1 as biological absence.

Table 17: Leading globally significant MapMan v3 terms in M5.

	Term	MapMan category	Hits	Fold	Global FDR
1	9.2.3	Stilbenoid biosynthesis	18	26.57	6.63e-23
2	9.2	Phenolics	25	11.68	4.87e-19
3	9	Secondary metabolism	25	9.60	1.04e-16
6	9.2.1.1.1	PAL	7	24.54	1.72e-07
7	9.2.1.1	PAL activity	7	21.81	6.43e-07
9	9.2.1	p-coumaroyl-CoA biosynthesis	7	17.85	4.32e-06

11.3 T-006: hub-gene prioritization

Intramodular connectivity was calculated on the frozen beta-10 adjacency. Eight of the 11 M5 top-decile hubs belong to the chromosome-16 family with conflicting STS/CHS labels. A copper amine oxidase candidate ranked fifth and a NAC-domain transcription-factor candidate ranked seventh; PAL genes were enriched at module level but were not in the top 11. M10 prioritized a bHLH candidate and a HAK/KUP/KT transporter. M2 prioritized MYB- and FAR1-labeled candidates, while recurrent exact zeros require mapping- and structure-aware interpretation.

Table 18: M5 top-decile intramodular hubs and selected annotation evidence.

Gene	Rank	kWithin	Annotation evidence
VIT_16s0100g00780	1	43.66	v3-STS v5-CHS
VIT_16s0100g00750	2	42.22	v3-STS v5-CHS
VIT_16s0100g00770	3	41.84	v3-STS v5-CHS
VIT_16s0100g01100	4	41.49	v3-STS v5-CHS
VIT_05s0020g03280	5	40.84	
VIT_16s0100g00910	6	40.77	v3-STS v5-CHS
VIT_12s0028g00860	7	40.28	NAC
VIT_16s0100g01140	8	40.20	v3-STS v5-CHS
VIT_06s0004g02010	9	39.32	
VIT_16s0100g01150	10	39.20	v3-STS v5-CHS
VIT_16s0100g01070	11	38.83	v3-STS v5-CHS

11.4 T-007: external skin-only expression support

GSE72421 and PRJNA260535 were analyzed as independent skin-only evidence, not as additional GSE98923 replicates. In GSE72421 well-watered skin, 11 of 11 evaluable M5 hubs passed BH correction within the predefined 37-hub set and matched the stable Harvest direction. In PRJNA260535 at 24 degrees Brix, four of seven evaluable M5 hubs passed. CuAO- and NAC-labeled M5 candidates agreed in both sources. M10 support was limited, whereas selected M2 MYB and FAR1 candidates agreed but had incomplete RNA-seq coverage.

Table 19: Evaluable and BH-supported predefined hubs in the two primary external comparisons.

Dataset	Condition	Module	Assayed	Significant
PRJNA260535	24	M10	3	0
GSE72421	WW	M10	4	1
PRJNA260535	24	M2	8	7
GSE72421	WW	M2	22	18
PRJNA260535	24	M5	7	4
GSE72421	WW	M5	11	11

11.5 T-008: modern raw-read processing and module preservation

All 54 FASTQ and Salmon outputs passed integrity and quantification checks. The modern matrix contains 47,971 genes and 54 samples. Preservation used only 1,922 audited reciprocal one-to-one legacy-to-v5.1 mappings and fixed historical memberships.

Table 20: Fixed-membership preservation against modern read reprocessing.

	Module	Genes	Mapped	Coverage	Eigengene r	Adjacency rho	Zsummary	Class
1	M1	2167	1483	68.4%	0.997	0.913	28.93	strong
6	M5	108	76	70.4%	0.986	0.827	7.49	moderate
2	M10	39	21	53.8%	0.996	0.878	3.74	moderate
3	M2	214	81	37.9%	0.996	0.775	7.45	moderate
4	M3	131	35	26.7%	0.420	0.319	-2.12	not_supported

M1 was strongly preserved. M5, M10 and M2 showed moderate preservation of their mapped cores with eigengene correlations of 0.986–0.998. M2 mapped only 81 of 214 genes, so the complete module is not declared preserved. M3 preservation was not supported.

12 Integrated interpretation

M5 has the strongest convergent evidence: the largest cultivar-by-stage interaction, a repeated Harvest direction across vintages, coherent phenolic-pathway enrichment, externally supported hubs

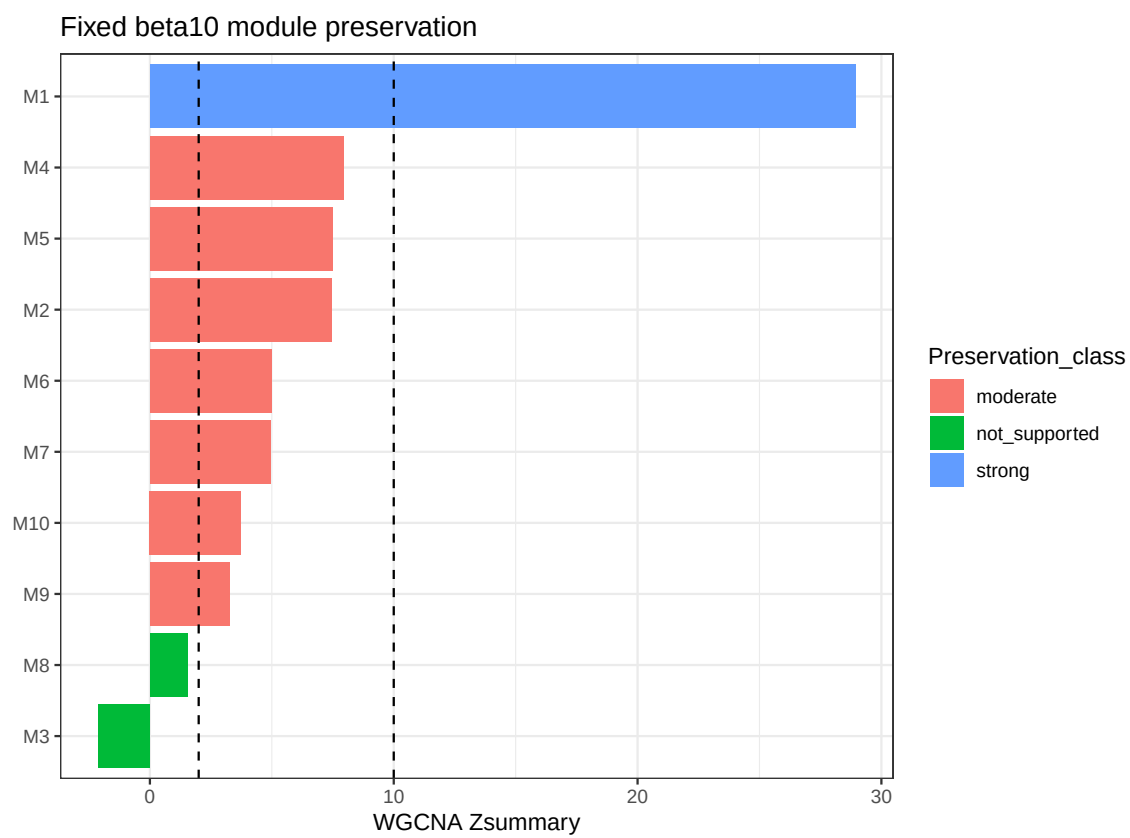


Figure 2: WGCNA module-preservation Zsummary statistics. Coverage and mapping ambiguity must be considered together with Zsummary.

and moderate mapped-core preservation after modern reprocessing. M10 has the strongest module-level vintage reproducibility but less functional and external support. M2 retains repeated Veraison and Harvest contrasts and a preserved mapped core, but limited mapping coverage and exact-zero patterns require structural and reference-aware follow-up. M3 and M1 remain lower-confidence biological priorities despite specific statistical or technical signals.

13 Final limitations

- The primary material is berry pericarp, not isolated skin.
- No analyzed dataset measures berry skin thickness.
- Cultivar is entangled with clone, rootstock and planting history in the original field system.
- External sources differ in year, condition, developmental definition and platform.
- MapMan and GO coverage are incomplete, and duplicated CHS/STS-family genes remain ambiguous.
- Reciprocal one-to-one mapping excludes ambiguous genes and therefore underrepresents some families.
- Coexpression, enrichment, hub status, external concordance and preservation are not causal evidence.

14 Final conclusion

The project supports cultivar-associated developmental transcriptional programs in grape berry pericarp. M5 is the leading integrated candidate program, M10 is the clearest year-reproducible module, and M2 warrants mapping- and structure-aware follow-up. No current result demonstrates that a module or gene causes thick or thin berry skin.